

CRD (Completely Randomized design)

o When experimental design is Homogeneous. (Laboratory / Greenhouse studies)

STEPS

1. Set Hypothesis.
2. Correction Factor.
3. TSS (Total sum of square)
4. TRSS (Treatment)
5. ESS (Error sum of square)
6. Anova
7. Conclusion

Step I: Set up Hypothesis

	Replications →					
	R ₁		R ₂		R ₃	
1	1.5		1.8		1.6	
2	4.7		4.4		5.3	
3	5.7		5.8		5.3	
4	3.7		3.2		2.9	
5	6.6		6.3		6	
6	5.8		6.1		5.5	

Step III: TSS
(Total sum of square)

Step IV: TRSS
(Treatment sum of square)

Play (k)

Step II: Correction Factor

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II

CC

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STEPS

1. Set Hypothesis.
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7. conclusion

Step I: Set up Hypothesis

$$H_0 = T_1 = T_2 = T_3 = T_4 = T_5 = T_6$$

$$H_1 = T_1 \neq T_2 \neq T_3 \neq T_4 \neq T_5 \neq T_6$$

Step II: Correction Factor

$$C.F = \frac{(G.T)^2}{18} = \frac{(82.2)^2}{18}$$

	R_1	R_1^2	R_2	R_2^2	R_3	R_3^2	Total
1	1.5	2.25	1.8	3.24	1.6	2.56	4.9
2	4.7	22.09	4.4	19.36	5.3	28.09	14.4
3	5.7	32.49	5.8	33.64	5.3	28.09	16.8
4	3.7	13.69	3.2	10.24	2.9	8.41	9.8
5	6.6	43.56	6.3	39.69	6	36	18.9
6	5.8	33.64	6.1	37.21	5.5	30.25	17.4
	28	147.72	27.6	143.38	26.6	133.4	82.2

Treatments

Step III: TSS
(Total sum of square)

$$\Rightarrow \frac{[147.72 + 143.38 + 133.4]}{3} - C.F$$

$$\Rightarrow \frac{424.5}{3} - \frac{47.82}{3}$$

$$\Rightarrow 141.5 - 15.94 = 125.56$$

Step IV: TrSS
(Treatment sum of square)

$$\Rightarrow \frac{4.9^2 + 14.4^2 + 16.8^2 + 9.8^2 + 18.9^2 + 17.4^2}{6} - C.F$$

$$\Rightarrow \frac{423.2}{6} - \frac{47.82}{6}$$

$$\Rightarrow 70.53 - 7.97 = 62.56$$

Step V :- ESS (Error sum of square)

$$ESS = TSS - TMS$$

$$49.12 - 47.82 = \underline{1.3}$$

Step VI :- Anova

Source of variation	df	SS	MSS	F ratio
Treatment	t-1	47.82		
Error	n-t	1.3		
Total	n-1	49.12		

Step VII :- Conclusion

Step V :- ESS (Error sum of square)

$$ESS = TSS - TMS$$

$$49.12 - 47.82 = \underline{1.3}$$

Step VI :- Anova

Source of variation	df	SS	MSS	F ratio
Treatment	$t-1$ $6-1=5$	47.82	$\frac{47.82}{5} = 9.56$	$\frac{9.56}{0.108}$
Error	$n-t$ $18-6=12$	1.3	$\frac{1.3}{12} = 0.108$	$\Rightarrow \underline{\underline{88.51}}$
Total	$n-1$ $18-1=17$	49.12		

Step VII :- Conclusion

STATISTICAL TABLES

Table 4. Significant Points of F (5 percent) with ν_1 and ν_2 degrees of freedom

$\nu_1 \backslash \nu_2$	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
1	161	200	216	225	230	234	237	239	241	242	244	246	248	249	250	251	252	253	254
2	18.5	19.0	19.2	19.2	19.3	19.3	19.4	19.4	19.4	19.4	19.4	19.4	19.4	19.5	19.5	19.5	19.5	19.5	19.5
3	10.1	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.36
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.02	2.97	2.93
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.84	2.77	2.74	2.70	2.66	2.62	2.58	2.54
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
12	4.75	3.89	3.49	3.26	3.10	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
13	4.67	3.81	3.41	3.18	3.02	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.21
14	4.60	3.74	3.34	3.11	2.95	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
15	4.54	3.68	3.29	3.06	2.90	2.80	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
16	4.49	3.63	3.24	3.01	2.85	2.75	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.10	2.06	2.01
17	4.45	3.59	3.20	2.96	2.80	2.70	2.61	2.54	2.49	2.44	2.38	2.31	2.23	2.19	2.15	2.11	2.06	2.02	1.97
18	4.41	3.55	3.16	2.93	2.77	2.67	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.11	2.07	2.03	1.98	1.93
19	4.38	3.52	3.13	2.90	2.74	2.64	2.55	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.99	1.95	1.90
20	4.35	3.49	3.10	2.87	2.71	2.61	2.52	2.45	2.39	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84
21	4.32	3.47	3.07	2.84	2.68	2.58	2.49	2.42	2.36	2.32	2.25	2.17	2.10	2.05	2.01	1.96	1.92	1.87	1.81
22	4.30	3.44	3.05	2.82	2.66	2.56	2.47	2.40	2.34	2.30	2.23	2.15	2.08	2.03	1.99	1.94	1.89	1.84	1.78
23	4.28	3.42	3.03	2.80	2.64	2.54	2.45	2.38	2.32	2.28	2.21	2.13	2.06	2.01	1.97	1.92	1.87	1.81	1.75
24	4.26	3.40	3.01	2.78	2.62	2.52	2.43	2.36	2.30	2.26	2.19	2.11	2.04	1.99	1.95	1.90	1.84	1.78	1.72

$$ESS = TSS - TMS$$

$$49.12 - 47.82 = \underline{1.3}$$

Step II :- Anova

Source of variation	df	SS	MSS	F ratio
Treatment	$t-1$ $6-1=5$	47.82	$\frac{47.82}{5} = 9.56$	$\frac{9.56}{0.108}$
Error	$n-t$ $18-6=12$	1.3	$\frac{1.3}{12} = 0.108$	$\Rightarrow \underline{88.51}$
Total	$n-1$ $18-1=17$	49.12		

Step III :- Conclusion

$$F_{Tab} = 3.11$$

$$F_{cal} = 88.51$$

$$F_{Tab} < F_{cal}$$

reject H₀



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